Parallel Trends and Methods-Related (Mis)communication in the Economics Literature*

Teresa Molina†

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Abstract

How well do users of empirical methodologies communicate their underlying assumptions? This paper examines this question in the context of the parallel trends assumption required for difference-in-differences (DiD) – the fundamentally untestable assumption that outcomes of treated and untreated units would have continued to evolve in parallel after the implementation of a treatment. Although a common way to shed light on the validity of this assumption is to examine pre-trends for treated and untreated units, parallel pre-trends are neither necessary nor sufficient for the validity of a DiD analysis. To examine how authors of economics publications navigate this nuance, I review all publications, in thirteen leading economics journals, that use the phrase "parallel trends" and examine pre-trends. I identify whether these papers (a) correctly state the identifying assumption required for a causal interpretation of their estimates and (b) accurately describe parallel pre-trends as support for (rather than confirmation of or a requirement for) the parallel trends assumption. Only two-thirds of papers do both, and 7% of papers do neither. Papers with fewer authors, references to other correct papers, and references to the modern DiD literature tend to have a lower degree of miscommunication. Based on citation counts, poor communication is not associated with lower impact.

Keywords: difference-in-differences, parallel trends

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†University of Hawaii at Manoa, IZA, UHERO; tmolina@hawaii.edu

1 Introduction

Causal inference methods that are easily implemented and widely applicable to different types of research questions grow popular rapidly. This is beneficial for knowledge creation because it enables more researchers to answer more questions using methods that are capable of isolating causal estimates. However, these methods are only able to identify causality under certain assumptions, which researchers must have the ability to (a) understand, (b) assess the validity of, and (c) communicate to readers. As it becomes increasingly easy to access statistical code written by others, it becomes increasingly easy to implement modern methods – even for those who lack formal training in or a deep understanding of the theory behind the methods, including the identifying assumptions.

How well do the implementers of econometric methods understand and communicate the assumptions required for them to be able to interpret their estimates causally? The answer to this question is important because it sheds light on the quality of research being generated. It also sheds light on whether current mechanisms for transmitting knowledge – specifically regarding new or adapted econometric methods – are working well. For certain methodologies, it is possible to obtain a clear picture of this by examining how authors write about specific nuanced, commonly misunderstood phenomena in their papers.

The parallel trends assumption required for the difference-in-differences (DiD) approach provides a compelling setting for this question. With the explosion of DiD papers and associated statistical code, the DiD methodology (often implemented using a two-way fixed effects regression) has become a dominant approach to answer causal questions, particularly those involving the evaluation of a policy or intervention that rolled out to different units (individuals, regions) at different times. Event-study figures, which plot relative trends in outcomes for treated and control units in the periods leading up to and after treatment implementation, are a common fixture in standard DiD papers.

All DiD papers require some form of a parallel trends assumption. In a simple set-up with one treatment and one control group, this requires that the treatment and control groups would have experienced similar trends in the outcome of interest, had the treatment group not received the treatment. Importantly, this assumption revolves around unobserved, counterfactual trends and therefore is fundamentally untestable: we do not know what would have happened to a treatment group in the absence of treatment. That said, diverging trends for the treated and control groups in the period leading up to treatment implementation would shed doubt on the validity of this assumption, which is why it is common for researchers to test whether treatment and control trends are similar in the periods prior to the treatment. Although analyzing pre-trends are often very useful, parallel pre-trends are neither necessary nor sufficient for the validity of the parallel trends assumption, a point that has been highlighted by related work (Baker et al., 2025; Kahn-Lang and

Lang, 2020; Roth, 2022).

This is an important nuance. There are settings in which violations of the parallel trends assumption are possible – sometimes, in fact, plausible – even when pre-trends are parallel. Focusing solely on pre-trends can shift focus away from important issues related to the counterfactual trends, which might require further examination using other variables or other tests. As highlighted by Roth (2022), conventional pre-trend tests may have low statistical power, and the practice of conditioning conducting a DiD analysis on the existence of similar pre-trends across treatment and control can induce distortions to estimation and inference. In short, it is crucial that implementers of the DiD methodology – and readers evaluating the validity of DiD results – understand this nuance.

In this paper, I examine how authors of economics publications navigate this particular issue. I examine over 300 publications that include the phrase "parallel trends" and a discussion of an event-study plot, published in the following journals: *American Economic Journal: Applied Economics* (AEJ Applied), *American Economic Journal: Economic Policy* (AEJ Policy), *American Economic Review* (AER), *Econometrica*, *Journal of Development Economics* (JDE), *Journal of Finance* (JF), *Journal of Health Economics* (JHE), *Journal of Labor Economics* (JoLE), *Journal of Political Economy* (JPE), *Journal of Public Economics* (JPubE), *Quarterly Journal of Economics* (QJE), *Review of Economic Studies* (REStud), and *Review of Economics and Statistics* (REStud). I examine how these paper discuss the parallel trends assumption and tests of pre-trends. First, I identify whether papers do what should be common practice in applied econometrics papers: correctly state the identifying assumption required for a causal interpretation of the estimates. I also identify whether papers discuss pre-trends accurately: that is, whether they describe parallel pre-trends as support for (rather than confirmation of) the parallel trends assumption. Only two-thirds of papers do both of these ("clear and correct"), and 7% of papers do neither ("incorrect"), meaning that they incorrectly state that parallel pre-trends confirm or are required for the parallel trends assumption. I classify the remaining 27% as "misinterpretable," either because they omit the identifying assumptions or because their wording is too strong when describing the implications of their pre-trend tests.

There are substantial differences across journals, but "incorrect" papers appear in all but three journals (Econometrica, JPE, and QJE). Papers are less likely to be classified as clear and correct and more likely to be classified as misinterpretable when there are three or more authors or when they also cite other misinterpretable papers. Papers are less likely to be incorrect when they cite papers from the modern DiD literature and when their abstracts are scored as more readable. Comparing citation counts across categories, I find no evidence that poor communication is associated with lower impact. I compare the largely manual categorization of papers used in the main analysis with an approach that relies more heavily on a large language model (LLM) and find similar (though not identical) results, indicating that LLMs are fundamentally

constrained by their (human) inputs.

This paper is related to a recent body of work exploring nuances related to pre-trend tests. The point that parallel pre-trends are not sufficient for a valid DiD strategy has been emphasized by other papers (Baker et al., 2025; Kahn-Lang and Lang, 2020). There are also other reasons to be cautious about pre-trend tests: they can have low power, and using parallel pre-trends as a screening device for DiD papers can introduce bias (Bilinski and Hatfield, 2023; Roth, 2022). Several papers offer alternative approaches to avoid these issues. For example, the approach of Rambachan and Roth (2023) involves relaxing the parallel trends assumption – allowing for differences up to a certain magnitude relative to the difference in pre-treatment trends – instead of relying on tests for similar pre-treatment trends. When a specific potential confounding factor can be identified (one that might generate differential pre-trends as well as violations of the parallel trends assumption), Freyaldenhoven et al. (2019) suggest using a covariate affected by the confounding factor but not by the treatment itself to correct for the confounder.

This paper also speaks to the literature on the quality of research in economics and other social sciences, which has touched upon replicability (Brodeur et al., 2026; Chang et al., 2018; Huntington-Klein et al., 2021), reporting errors (Pütz and Bruns, 2021), and misleading causal claims (Isch et al., 2026). A related strand of work examines the determinants and consequences of writing style in economics. For example, correlational evidence from McCannon (2019) documents that more readable papers are cited more. Hengel (2022) finds that average readability scores of economics publications differ by author gender, which she shows is an indication that the field imposes higher writing standards for women. Also interested in gender differences, Kosnik (2022) documents that men write with more positive sentiment than women. Focusing on a different aspect of economic writing, Garg and Fetzer (2026) show that causal claims in economics have increased over time, and that having a causal narrative is correlated with better publication outcomes. This paper examines a characteristic that is both an element of writing style and a measure of quality: the precision with which authors describe their identifying assumptions.

2 The Parallel Trends Assumption in DiD Analysis

The classic 2x2 DiD framework consists of two groups of units indexed by i and two time periods, before ($t = 0$) and after ($t = 1$) the implementation of a treatment to a subset of the units. The treated group ($D_i = 1$) is untreated in $t = 0$ but treated in $t = 1$. The control group ($D_i = 0$) is untreated in both periods. I use a potential outcomes framework where $Y_{i,t}(1)$ denotes the potential outcome for unit i at time t in a state of the world where it receives treatment in $t = 1$, and $Y_{i,t}(0)$ denotes unit i's potential outcome at time

t in a state of the world where it remains untreated in both periods.[1] The target parameter is the Average Treatment Effect on the Treated (ATT) in the post-period ($t = 1$):

$$ATT = \mathbb{E}[Y_{i,1}(1) - Y_{i,1}(0)|D_i = 1] = \mathbb{E}[Y_{i,1}(1)|D_i = 1] - \mathbb{E}[Y_{i,1}(0)|D_i = 1]. \tag{1}$$

One key issue is that we can only observe one of the two potential outcomes for each group. That is, the observed outcome $Y_{i,t}$ is equal to

$$Y_{i,t} = D_i Y_{i,t}(1) + (1 - D_i)Y_{i,t}(0),$$

which means we do not observe the second term in (1). However, if we assume that, had the treatment had not occurred, the change in the average outcome among the treatment group would have been the same as the change in the average outcome among the control group, this helps us recover the ATT. This assumption, referred to as the "parallel trends assumption" can be expressed as follows:

$$\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0)|D_i = 1] = \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0)|D_i = 0]. \tag{2}$$

Together with the "no anticipation" assumption that treatment does not have any effect on the treatment group prior to $t = 1$ (i.e., that $Y_{i,0}(0) = Y_{i,0}(1)$ for individuals in the treated group), this allows us to rearrange the expression in (2) and obtain the unobserved component of the ATT in (1).

$$\begin{aligned}
\mathbb{E}[Y_{i,1}(0)|D_i = 1] &= \mathbb{E}[Y_{i,0}(0)|D_i = 1] + \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0)|D_i = 0] \\
\mathbb{E}[Y_{i,1}(0)|D_i = 1] &= \mathbb{E}[Y_{i,0}(1)|D_i = 1] + \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0)|D_i = 0] \\
\mathbb{E}[Y_{i,1}(0)|D_i = 1] &= \mathbb{E}[Y_{i,0}|D_i = 1] + \mathbb{E}[Y_{i,1} - Y_{i,0}|D_i = 0].
\end{aligned}$$

Substituting this into (1), the ATT can be calculated as:

$$ATT = \mathbb{E}[Y_{i,1} - Y_{i,0}|D_i = 1] - \mathbb{E}[Y_{i,1} - Y_{i,0}|D_i = 0]. \tag{3}$$

In practice, this is often estimated via a Two-Way Fixed Effects (TWFE) regression:

$$Y_{i,t} = \alpha_i + \phi_t + \beta T_{it} + \epsilon_{i,t}$$

[1] Baker et al. (2025) use this as shorthand notation for $Y_{i,t}(0,1)$ and $Y_{i,t}(0,0)$, respectively, where the numbers in parentheses represent the full treatment status trajectory.

where T_{it} represents a time-varying indicator equal to 1 if i is treated in time t. Under the parallel trends and no anticipation assumptions, $\hat{\beta}$ – the DiD estimate – provides a consistent estimate of the ATT.

Equation (2) makes it clear that the parallel trends assumption hinges on counterfactual trends: researchers do not observe $\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0)|D_i = 1]$ – treatment group trends in the absence of treatment. They must therefore provide convincing evidence that the observed control group trends ($\mathbb{E}[Y_{i,1} - Y_{i,0}|D_i = 0]$) are a reasonable proxy for them. In situations where researchers have access to multiple pre-treatment periods (rather than just $t = 0$ in the simple two-period setting above), one common way to provide evidence for this assumption is to compare treatment and control trends across periods for $t < 0$. The existence of pre-trends that evolve in parallel across treatment and control is often cited as support for the assumption that they would have continued to trend in parallel had the treatment not been implemented.

While parallel pre-trends can certainly lend support to the validity of the parallel trends assumption, they can never confirm or prove it. Consider the example of policy that is implemented only in some regions of a country. If initial analysis reveals that characteristics of treated regions were systematically different from characteristics of untreated regions before the policy was implemented, one might be concerned that their outcomes were already following different trajectories. In this case, showing parallel pre-trends would help alleviate concerns about the validity of the DiD strategy because it would indicate that the groups were trending similarly despite their baseline differences (and would have likely continued to do so in the absence of the policy). However, policies are rarely implemented in isolation. If the policy took place among other regional changes (including other government policies or changes in regional populations), then the parallel trends assumption may still be violated even if pre-trends appear identical.

Some DiD strategies exploit pre-existing variation in how much a nationwide shock affects different units. Consider the case of the eradication of a disease. While the eradication might technically affect the entire country at a single point in time, its expected impact is larger in regions where the disease was endemic compared to regions where the disease was less common, prior to eradication. It is therefore possible to recover the effect of disease eradication using a DiD analysis, under the assumption that outcomes in endemic and non-endemic regions would have evolved in parallel in the absence of eradication. If endemic regions were systematically poorer than non-endemic regions before eradication, one might worry that their outcomes were already on different trajectories. Demonstrating parallel pre-eradication trends would help alleviate this concern. Even so, the details of the nationwide shock matter: if the eradication was brought about by an external factor, like a medical breakthrough abroad, then parallel pre-trends might be sufficiently convincing. However, if the eradication was a result of domestic improvements in medical science research, these improvements might have differentially affected low-income regions even if the disease had not been successfully eradicated. In this case, parallel pre-trends might not mean that counterfactual trends would

have remained parallel: low-income regions might have exhibited different trends in outcomes due to medical research investments, even without the successful eradication of the disease.

In practice, there are additional concerns associated with testing for significantly different pre-trends across treatment and control groups. Pre-trend tests may suffer from low power, which means they can fail to reject the null of parallel pre-trends even when violations of the parallel trends assumption do exist. Moreover, using parallel pre-trends as a screening device for publication has ambiguous effects on the average bias among papers that are eventually published (Roth, 2022).

In addition to being not sufficient, parallel pre-trends are also not necessary for a valid DiD, although cases where non-parallel pre-trends exist despite the counterfactual trends being parallel are likely rare. As pointed out by Baker et al. (2025), pre-trends that are measured too far in advance of the treatment implementation date may not be very informative about the plausibility of parallel counterfactual trends after treatment begins. Moreover, short-term, temporary shocks to treatment units in the pre-period might generate differential pre-trends without affecting counterfactual post-treatment trends. In some cases, there are mechanical reasons for differential pre-trends that do not threaten the validity of the parallel trends assumption (Deshpande, 2016).

This paper will focus on how researchers describe their analyses of pre-trends. While parallel pre-trends might often coincide with the validity of the parallel trends assumption, we cannot know how frequently this is true or correctly identify when this is the case because the counterfactual state is unobservable. It is therefore incorrect to say that parallel pre-trends prove or confirm the parallel trends assumption holds. How well researchers understand and communicate this nuance is important. DiD strategies are often used for program evaluation questions intended to inform policy, which means the conclusions from these studies often have far-reaching implications. An over-reliance on pre-trend testing can distract researchers from more substantive threats to identification. In the regional policy example outlined above, a researcher who stops at a satisfactory pre-trend test might overlook contemporaneous policy shifts that – if properly accounted for – would yield a significantly different and more accurate estimate of the program's true effect.

It is important to acknowledge that recent empirical work rarely uses the simple 2x2 framework described above. In practice, researchers typically examine multiple time periods, multiple treatment groups, and treatments that roll out to different units at different times. A growing literature has shown that these more complicated DiD approaches require stronger and more precisely defined parallel trends assumptions than the simple 2x2 version described above.[2] For example, when treatment adoption is staggered, researchers often rely on TWFE regressions, which implicitly require an assumption of homogeneous treatment effects over time and across cohorts to identify an interpretable average treatment effect. In the presence of heterogeneous

[2]See Baker et al. (2025) and Roth et al. (2023) for reviews.

treatment effects (across units treated at different times), these estimators can perform "forbidden comparisons" that use already-treated units as controls for later-treated units, potentially resulting in negative weights and incorrectly signed estimates (Borusyak et al., 2024; de Chaisemartin and D'Haultfœuille, 2023; Goodman-Bacon, 2021; Sun and Abraham, 2021). Furthermore, when analyzing continuous or multi-valued treatments, identifying causal response parameters typically requires a "strong" parallel trends assumption that holds for potential outcomes across all possible treatment levels, rather than just for the never-treated group (Callaway et al., 2024). Finally, in designs where the parallel trends assumption is only plausible after accounting for observed unit characteristics, researchers must invoke a conditional parallel trends assumption, which requires that the path of untreated outcomes is the same only for units sharing the same covariate values (Caetano and Callaway, 2023; Callaway and Sant'Anna, 2021).

These additional complications create numerous opportunities for researchers to potentially misunderstand or poorly communicate the necessary identifying assumptions for their empirical approaches. However, this paper ignores these additional issues and focuses specifically on the narrow misconception that parallel pre-trends serve as a necessary or sufficient condition for the validity of a DiD strategy. Given this narrow focus, the findings presented here provide a conservative estimate of the total magnitude of methods-related miscommunication in DiD papers.

3 Dataset Construction

Because the goal of this paper is to examine how authors discuss the parallel trends assumption and describe empirical tests of pre-trends, the first step of dataset construction involved a Google Scholar search for all articles with both of the phrases "event study" and "parallel trends." I searched in what is commonly referred to as the "top 5" economics journals (AER, Econometrica, JPE, QJE, and REStud) or any of the following journals, which commonly publish papers featuring event studies: AEJ Applied, AEJ Policy, JDE, JF, JHE, JoLE, JPubE, or REStat.[3] After dropping papers that were purely methodological (e.g., developing a new DiD-related estimator) or that did not discuss pre-trends as part of a standard event study analysis, a total of 335 publications formed my final sample, with the earliest paper published in 2014; I use 2025 as the end year cutoff. While DiD papers certainly existed before this, analyses of pre-trends were less common. That said, this sample likely excludes a number of potentially relevant papers (both before and during the period 2014-2025) that do not use the phrase "event study" but still conduct an analysis of pre-trends.

[3]With a few exceptions, these journals ranked in the top 25 of IDEAS RePEc's Aggregate Rankings (based on data from the past 10 years) and published at least 10 papers with the two focal phrases by 2025. Journals like JHE and JPuBE fell outside of the top 25 but were included due to the large number of papers (30 or more) satisfying the criteria.

3.1 Paper Classification

For each paper, I used Google NotebookLM to extract all instances of the phrases "identifying assumption," "identification assumption," "parallel trends," and "pre-trends." Using the extracted sentences, I determined whether the paper correctly described the identifying assumption required for the difference-in-differences analysis in the paper. If this could not be determined from the selected phrases, I went back to the full paper and skimmed for relevant phrases to make this decision. Because a key question of interest is whether a reader could misconstrue the phrase "parallel trends" as referring to pre-trends rather than counterfactual trends, papers were not classified as correct if their discussion of identifying assumptions was vague (for example, stating that identification relied on the assumption that "treatment and control were on parallel trends" or simply "parallel trends between treatment and control.")[4] I then classified how the paper discusses the results of pre-trend analysis: do they state or imply that parallel pre-trends *support* the parallel trends assumption (which is usually correct), or do they state or imply that parallel pre-trends *confirm* or are *required* for the parallel trends assumption (which is always incorrect)?[5] Although the difference between the two is subtle, papers that do the latter are making an incorrect statement because the parallel trends assumption is fundamentally untestable and nothing, including parallel pre-treatment trends, can confirm or definitively validate it. I define these categories to be mutually exclusive: papers can discuss pre-trends multiple times in different ways, but any incorrect statement about pre-trends proving the validity of the parallel trends assumption will place a paper in the second category.

Cross-tabulating these two characteristics (how papers describe the identifying assumptions and how they discuss pre-trends) creates four different categories of papers, as summarized in Table 1. Papers that correctly state their identifying assumptions and correctly describe the implications of their pre-trend analysis are classified as "clear and correct." These papers leave little room for confusion about what the required identifying assumptions are and that they hinge on counterfactual trends as opposed to pre-trends. Papers that do not do either of these two things are classified as "misinterpretable," either because their wording is too strong (they claim that pre-trends can prove the parallel trends assumption) or because they omit any discussion of identifying assumptions. If a paper fell into any category other than "clear and correct" after my initial read of the sentences extracted by NotebookLM, I went back to the paper to seek out descriptions of the identifying assumption and re-categorized if necessary.

[4]The identifying assumption is not always stated in terms of counterfactual trends. Several papers described the identifying assumption in terms of exogeneity of the treatment variable, conditional on fixed effects, for example, which could still result in confusion for inexperienced readers who also see a discussion of parallel pre-treatment trends as support for the parallel trends assumption. Nevertheless, these papers are classified as correctly describing the identifying assumption.

[5]Hypothetical examples of the former include "pre-trends are parallel prior to treatment, providing support for the parallel trends assumption," or "there are no differential pre-trends prior to treatment, suggesting that counterfactual trends would have trended in parallel after the treatment." Hypothetical examples of the latter include "pre-trends are parallel prior to treatment, confirming that the parallel trends assumption holds," or "the key identifying assumption is that treatment and control were on parallel trends in the periods leading up to the treatment."

Table 1: Definition of Paper Categories

	Correctly describes identifying assumption	Does not describe identifying assumption
Parallel pre-trends *support* validity of DiD	Clear and correct	Misinterpretable (omits identifying assumptions)
Parallel pre-trends *confirm* validity of DiD	Misinterpretable (wording too strong)	Incorrect

3.2 Other Variables

From citation information, I extracted the number of authors and number of pages (where available) for each publication. Article length was calculated as a standardized measure: subtracting the journal-specific mean and dividing by the journal-specific standard deviation page counts.[6] Using the IDEAS RePEc ranking of Economics departments as of July 2026, I generate a dummy equal to 1 for papers with at least one author from a top 20 institution.

I also obtained information on references from OpenAlex, which provides the number of references as well as a list of references for each publication. In addition to the total number of references, I am interested in whether papers are more likely to be classified in a certain category if they cite other papers of that same category. I therefore identify papers that reference other papers in my sample and count the total number of "Correct," "Misinterpretable," and "Incorrect" references. 31% of the sample publications cite at least one other paper in the sample. OpenAlex also provides the number of citations for each paper.

In addition, I create a list of "modern" DiD papers using the references in the Baker et al. (2025) review paper. From this reference list, I include all methodological papers, dated no earlier than 2018, which mention difference-in-differences, two-way fixed effects, staggered rollout, event study, parallel trends, or pre-trends in the title or abstract. I generate a list of OpenAlex ID's for all of these papers (and any previous working papers, for those that are already published) and generate an indicator variable equal to 1 for papers in my sample that cite at least one modern DiD paper from this list.

Finally, I calculate the readability score of the abstract of each paper in the sample. I use the Gunning Fog Index, designed to estimate the number of years of formal education required to understand a piece of text on first reading. Because higher values of this index capture less readable text, I multiply by -1 so that higher values represent better readability. Table 2 reports summary statistics for the 335 sample papers.

[6] This information was not available in the citations pulled from JDE, JHE, and JPubE, for which page counts were manually recorded instead.

Table 2: Summary Statistics

	Mean	Std. Dev.	Min	Max
N Pages	27.31	12.67	5.00	62.00
Article Length	-0.02	0.94	-3.82	3.20
Top 20 Department	0.36	0.48	0.00	1.00
N References	50.93	19.11	1.00	129.00
N Incorrect Refs	0.01	0.11	0.00	1.00
N Misinterpretable Refs	0.09	0.29	0.00	1.00
N Correct Refs	0.32	0.63	0.00	4.00
Any New DiD Refs	0.45	0.50	0.00	1.00
Readability	-18.92	2.48	-25.83	-12.91
N Citations	59.57	126.81	0.00	1385.00

Notes: Total number of papers = 335. Article Length is the number of pages, standardized using the journal-specific mean and standard deviation. Readability is the Gunning Fog Index, multiplied by −1 so that higher values represent higher readability.

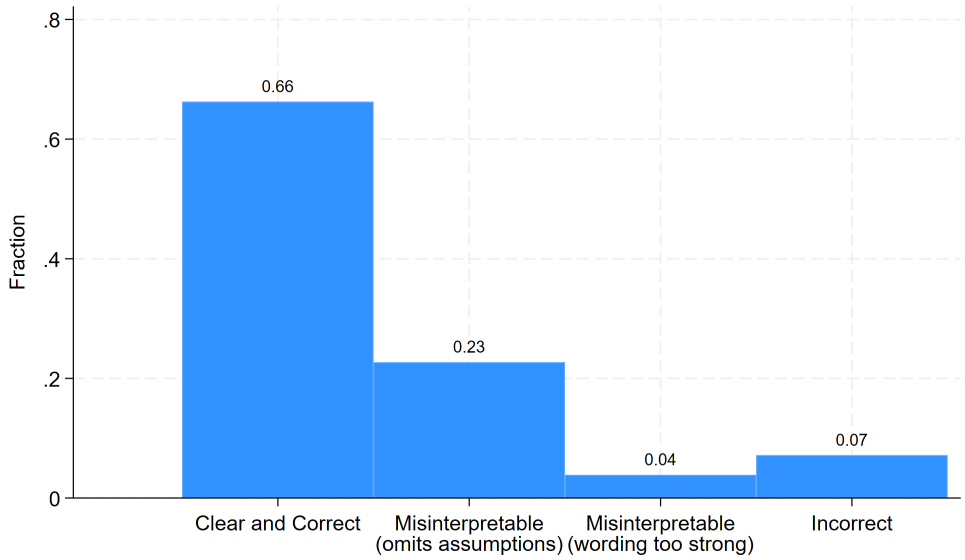
4 Results

4.1 Main Classification Results

As documented in Figure 1, approximately two-thirds of papers (66%) were classified as "clear and correct" – that is, they stated the identifying assumptions and discussed parallel pre-trends as support for (and not confirmation of) the parallel trends assumption. "Misinterpretable" papers comprised 27% of the sample, the vast majority of which were misinterpretable because they omitted the identifying assumptions. Finally, 7% of all papers made an incorrect statement about pre-trends without providing a correct description of the identifying assumptions and were therefore classified as "Incorrect." Only 70% correctly stated the identifying assumptions (the sum of the left two categories), and a total of 11% made an incorrect statement about parallel pre-trends (the sum of the right two categories). These results indicate a non-trivial amount of miscommunication regarding parallel trends, and there are two key reasons why they likely underestimate the extent of miscommunication that exists in the broader economic literature. First, the journals in the sample are among the best in the field – extremely selective and involving rigorous peer review. Second, the sample only includes papers that use the phrase "parallel trends" (in order to focus specifically on how papers discuss this assumption). Therefore, the share of papers that discuss identifying assumptions is likely to be higher in this sample than a broader sample of difference-in-differences papers, even those published in the same journals.

Figure 2 shows the evolution of paper counts and categorizations over time. The number of papers that use event studies and discuss pre-trends has risen dramatically over time. The paper classification distribution

Figure 1: Paper Classification



Paper categorization definitions described in Figure 1.

shifts year-to-year, though these shifts do not demonstrate any clear time trend in the shares of any category (which is shown more clearly in the regressions).

Figure 3 display differences across journals. Journals differ in the number of event study publications in the sample, ranging from only 4 for QJE to 60 for AEJ Policy. They also differ in the distribution of papers across categories. While "top 5" journals do appear to have smaller shares of "Incorrect" papers and larger shares of "Correct" papers, "Incorrect" papers do show up in top 5 journals like AER and REStud.

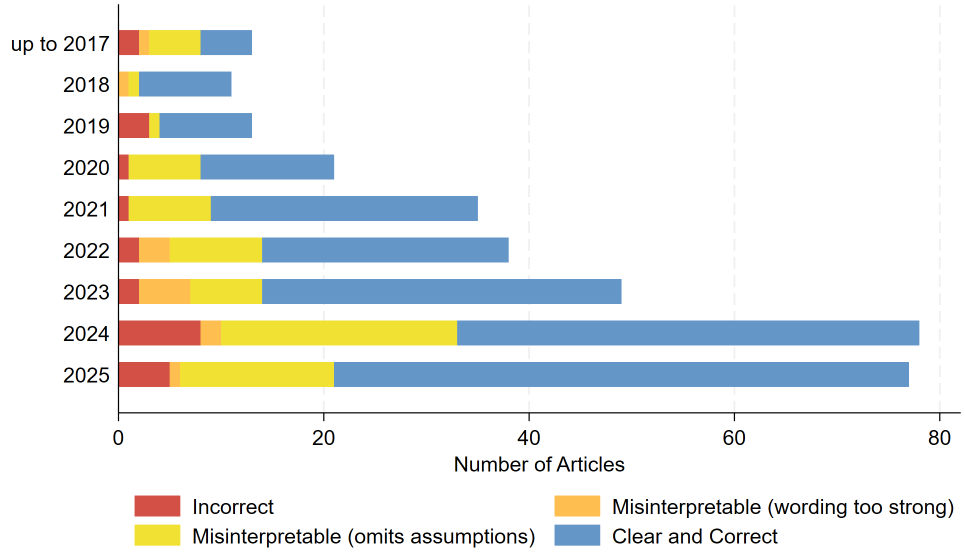
To examine the correlates of paper classification, I run the following regression:

$$Y_i = \beta X_i + \mu_{j(i)} + \delta_{y(i)} + \epsilon_i. \tag{4}$$

Y_i represents one of three dummy variables for paper i (which was published in journal j in year y), indicating its classification as "incorrect," "clear and correct," or "misinterpretable" (pooling both misinterpretable categories). In the basic specification, X_i represents a vector of paper-specific controls: indicator variables for papers with 2 authors, 3 authors, and 4 or more authors (leaving single-authored papers as the omitted category), paper length (both a linear and quadratic term), and total number of references cited by the paper (both a linear and quadratic term).

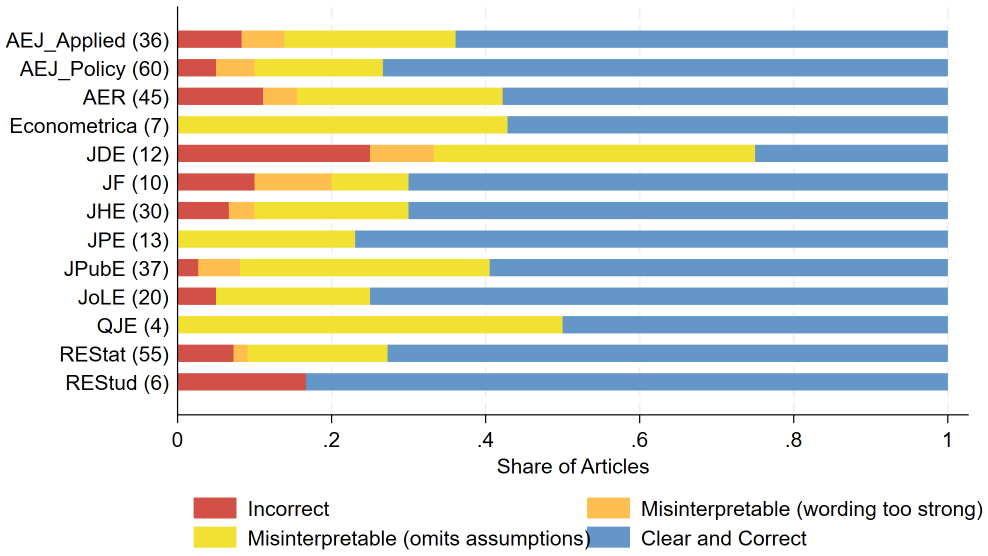
Table 3 reports the results of a simpler version of equation (4), which only includes journal and year fixed

Figure 2: Paper Classification Over Time



Paper categorization definitions described in Figure 1.

Figure 3: Paper Classification by Journal



Paper categorization definitions described in Figure 1. Numbers in parentheses represent total number of articles per journal.

effects. Results echo the findings of Figures 2 and 3. Relative to QJE (the omitted category), the journals AEJ Applied, AER, JDE, and REStat are significantly more likely to have "Incorrect" articles. Relative to 2025, there were significantly fewer "Correct" papers and significantly more "Misinterpretable" papers in the years 2014-2017, but after this, there does not seem to be a clear time trend in any of the outcome variables. Although certain years appear better than others, improvement or declines have not been monotonic.

Table 4 adds paper-specific controls to the regression. Authorship coefficients reveal clear patterns. Papers with four or more authors are significantly less likely to be classified "Correct" and more likely to be classified as "Misinterpretable." While more authors could mean more scrutiny and proof-reading, it could also mean that each coauthor pays less attention to ensuring that details about identifying assumptions are discussed. Article length appears largely uncorrelated with paper classification, as is the total number of references cited by a paper. However, the types of references matter. Papers that cite more "Misinterpretable" papers are significantly more likely to be classified as "Misinterpretable" themselves (and less likely to be classified as "Correct"), suggesting the transmission of knowledge or writing styles through citation networks. Relatedly, more "Correct" references are associated with a lower likelihood of being classified as "Incorrect." Papers that cite at least one modern DiD reference are more likely to be "Correct" and less likely to be "Incorrect," suggesting that authors who are familiar with the recent DiD literature, which has generally placed greater scrutiny on identifying assumptions, are able to communicate their own assumptions more clearly and accurately. Papers with abstracts scored as being more readable are also less likely to be classified as "Incorrect," which could reflect authors who place more effort into clear communication also being more careful about discussing identifying assumptions. Having an author in a top 20 department has no significant effects on paper classification.[7]

Having explored the correlates of paper classification, I now shift attention to the citation outcomes of different paper categories. To do this, I estimate the following regression:

$$Y_i = \alpha_1 1(\text{Misinterpretable})_i + \alpha_2 1(\text{Incorrect})_i + \beta X_i + \mu_{j(i)} + \delta_{y(i)} + \epsilon_i, \tag{5}$$

where Y_i represents a citation-related outcome: either the number of citations (for which I use PPML) or a dummy variable for having citations in the top half or top quartile of the distribution (for which I use OLS). The α_1 and α_2 coefficients can be interpreted as the difference in Y_i relative to the omitted "Correct" category.

In Table 5, I begin with a specification with no controls, add year and journal fixed effects, then add the full set of controls used above. Across all specifications, I find no significant coefficients on either the

[7] I reach the same conclusion with other ranking cutoffs – top 10, 30, 50, and 100.

Table 3: Paper Classification by Journal and Year

	(1) Correct	(2) Misinterpretable	(3) Incorrect
AEJ_Applied	0.08	-0.17	0.09*
	(0.28)	(0.27)	(0.05)
AEJ_Policy	0.17	-0.23	0.06
	(0.27)	(0.26)	(0.04)
AER	0.04	-0.17	0.13**
	(0.28)	(0.26)	(0.06)
Econometrica	-0.02	-0.01	0.02
	(0.32)	(0.32)	(0.04)
JDE	-0.32	0.06	0.26**
	(0.30)	(0.30)	(0.13)
JF	0.14	-0.28	0.14
	(0.31)	(0.28)	(0.10)
JHE	0.14	-0.22	0.08
	(0.28)	(0.27)	(0.05)
JPE	0.20	-0.23	0.03
	(0.29)	(0.28)	(0.03)
JPubE	0.01	-0.05	0.04
	(0.28)	(0.27)	(0.04)
JoLE	0.17	-0.22	0.05
	(0.28)	(0.27)	(0.05)
REStat	0.17	-0.26	0.09*
	(0.27)	(0.26)	(0.05)
REStud	0.25	-0.45*	0.20
	(0.32)	(0.26)	(0.16)
up to 2017	-0.36**	0.28*	0.08
	(0.14)	(0.15)	(0.11)
2018	0.04	0.01	-0.05*
	(0.13)	(0.13)	(0.03)
2019	-0.07	-0.11	0.18
	(0.14)	(0.09)	(0.12)
2020	-0.12	0.15	-0.03
	(0.13)	(0.12)	(0.06)
2021	0.02	0.03	-0.04
	(0.09)	(0.09)	(0.04)
2022	-0.14	0.15*	-0.01
	(0.09)	(0.09)	(0.05)
2023	-0.03	0.06	-0.03
	(0.08)	(0.08)	(0.04)
2024	-0.14*	0.10	0.04
	(0.08)	(0.07)	(0.04)
Observations	335	335	335
Outcome mean	0.66	0.27	0.07

Robust standard errors reported in parentheses (* p< 0.1, ** p< 0.05, *** p< 0.01). Paper categorization definitions described in Figure 1. Omitted categories are QJE (for journals) and 2025 (for years).

Table 4: Correlates of Paper Classification

	(1)	(2)	(3)
	Correct	Misinterpretable	Incorrect
2 authors	-0.12*	0.10	0.03
	(0.07)	(0.07)	(0.04)
3 authors	-0.10	0.05	0.05
	(0.08)	(0.07)	(0.04)
4 or more authors	-0.28***	0.25***	0.03
	(0.08)	(0.09)	(0.04)
Article Length	0.01	-0.01	0.00
	(0.03)	(0.03)	(0.01)
Article Length^2	-0.00	0.01	-0.01
	(0.02)	(0.02)	(0.01)
N References	0.02	-0.01	-0.00
	(0.04)	(0.04)	(0.02)
N References^2	-0.03*	0.02	0.01
	(0.02)	(0.02)	(0.01)
N Incorrect Refs	-0.02	-0.01	0.03
	(0.02)	(0.02)	(0.03)
N Misinterpretable Refs	-0.05*	0.07***	-0.02**
	(0.03)	(0.03)	(0.01)
N Correct Refs	0.03	-0.01	-0.02**
	(0.03)	(0.03)	(0.01)
Any New DiD Refs	0.12*	-0.01	-0.11***
	(0.06)	(0.06)	(0.03)
Readability	0.02	0.02	-0.03**
	(0.03)	(0.03)	(0.02)
Top 20 Department	-0.04	0.01	0.03
	(0.06)	(0.06)	(0.03)
Observations	335	335	335
Outcome mean	0.66	0.27	0.07

Robust standard errors reported in parentheses (* p< 0.1, ** p< 0.05, *** p< 0.01). Paper categorization definitions described in Figure 1. All regressions control for year and journal fixed effects.

"Misinterpretable" or "Incorrect" dummy variables. That is, "Misinterpretable" or "Incorrect" papers are no less likely to be cited than "Correct" papers. Although coefficients are large and positive in the first three columns (for number of citations), most of the remaining coefficients are negative, suggesting no clear pattern.

Table 5: Citations and Paper Classification

	(1) N Citations	(2) N Citations	(3) N Citations	(4) Citations ≥ 50th perc.	(5) Citations ≥ 50th perc.	(6) Citations ≥ 50th perc.	(7) Citations ≥ 75th perc.	(8) Citations ≥ 75th perc.	(9) Citations ≥ 75th perc.
Misinterpretable	0.42	0.21	0.15	-0.01	-0.03	-0.03	0.01	-0.01	-0.03
	(0.29)	(0.19)	(0.18)	(0.06)	(0.06)	(0.06)	(0.06)	(0.05)	(0.05)
Incorrect	0.53	0.09	-0.07	-0.01	-0.03	-0.02	-0.00	-0.04	-0.07
	(0.42)	(0.24)	(0.22)	(0.11)	(0.10)	(0.10)	(0.09)	(0.07)	(0.08)
2 authors			0.32*			0.03			0.02
			(0.17)			(0.06)			(0.06)
3 authors			0.00			-0.07			0.00
			(0.22)			(0.07)			(0.06)
4 authors			0.35			0.04			0.09
			(0.22)			(0.08)			(0.07)
Article Length			-0.01			0.03			-0.01
			(0.08)			(0.03)			(0.02)
Article Length^2			-0.10**			-0.02			-0.01
			(0.04)			(0.02)			(0.01)
N References			-0.16			-0.07**			0.01
			(0.12)			(0.03)			(0.03)
N References^2			0.09***			0.01			0.00
			(0.03)			(0.01)			(0.01)
N Incorrect Refs			-0.04			-0.00			0.02
			(0.04)			(0.03)			(0.02)
N Misinterpretable Refs			0.08			0.02			0.02
			(0.08)			(0.03)			(0.02)
N Correct Refs			-0.00			0.03			0.01
			(0.09)			(0.03)			(0.03)
Any New DiD Refs			0.11			0.10*			-0.01
			(0.21)			(0.06)			(0.05)
Readability			0.07			0.01			0.00
			(0.06)			(0.02)			(0.02)
Top 20 Department			0.29*			0.12**			0.10*
			(0.16)			(0.06)			(0.05)
Observations	335	335	335	335	335	335	335	335	335
Outcome mean	59.57	59.57	59.57	0.50	0.50	0.50	0.25	0.25	0.25
Fixed Effects	None	Year and Journal	Year and Journal	None	Year and Journal	Year and Journal	None	Year and Journal	Year and Journal

Robust standard errors reported in parentheses (* p< 0.1, ** p< 0.05, *** p< 0.01). Paper categorization definitions described in Figure 1. Columns 1 to 3 estimated using PPML.

4.2 LLM Classification Results

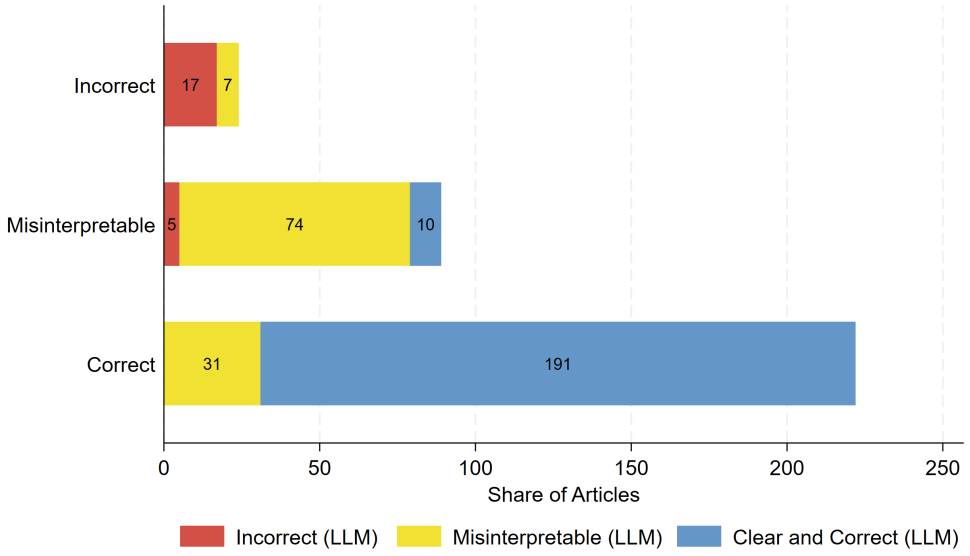
In addition to the largely manual categorization procedure described above, I also explored an alternative classification approach that relied more heavily on an LLM. Specifically, I asked Claude Code (Opus 5.5 model) to answer the following question for each paper in the sample: "Based only on the information in

this source, explain the parallel trends assumption. Make sure to ONLY use information from the source, nothing else. No internet, no information from other articles." This resulted in a few sentences of text for each paper. Based on these snippets of text, I categorized each paper as "Correct," "Incorrect," or "Misinterpretable," where the latter grouped those with wording that was too strong and those that omitted identifying assumptions.

There are several motivations for this analysis. LLMs like Claude Code operate as black boxes, and there is nothing to guarantee that their answer to the above question will provide an accurate representation of the underlying paper (or that they will indeed base their answer solely on the specified source, as requested). However, a comparison of the manual and LLM-based classification approaches does provide a rough sanity check for a classification procedure that relied on the selective reading of each paper (selecting phrases around key words and only going back to the text if more information was needed). Reassuringly, as shown in Figure 4, there is a large degree of similarity between the manual and LLM classifications. More specifically, 71% of manually-classified "Incorrect" papers were classified in the same way using the LLM-based approach. These figures were even higher for "Correct" (86%) and "Misinterpretable" (83%) papers. Discrepancies between the manual and LLM-based classifications operated in both directions: that is, the LLM-based classification was more generous in some instances (marking some manually-determined "Incorrect" papers as "Misinterpretable (LLM)" and manually-determined "Misinterpretable" papers as "Correct (LLM)") and harsher in others (classifying manually-determined "Misinterpretable" papers as "Incorrect (LLM)" and manually-determined "Correct" papers as "Misinterpretable (LLM)") This results in the overall distribution being similar, with 7% of papers classified as "Incorrect (LLM)" (the same as in the original classification), 33% classified as "Misinterpretable (LLM)" (slightly more than in the original classification) and 60% classified as "Correct (LLM)" (slightly less than in the original classification). Explorations of the correlates of this classification (Table A1) and differences in citation counts across these new categories (Table A2) yield similar patterns as those in Tables 4 and 5.

It is important to note, however, that the two exercises are fundamentally different. The first was a more open-ended exercise that explored how authors described the parallel trends assumption and their analysis of pre-trends. The second exercise explicitly asked for an an explanation of the parallel trends assumption (although the LLM output typically included additional information – for example, details about the identifying assumption used in the paper if the parallel trends assumption was not explicitly defined). In a word where researchers are becoming increasingly reliant on LLMs for synthesizing information, this exercise reveals how researchers who rely on LLMs to understand the parallel trends assumption will have that assumption presented to them. While researchers typically rely on multiple sources and do not restrict themselves to a single source at a time (as I asked Claude Code to do in this exercise), these results provide

Figure 4: Manual vs. LLM-Based Classification



evidence that inaccuracies and ambiguities do end up being transmitted through LLMs.

5 Discussion

This analysis reveals a surprisingly high degree of misrepresentation of the parallel trends assumption, especially considering the high caliber of the journals used for the analysis. One could argue that the authors themselves fully understand what the parallel trends assumption is but lack clarity in their discussion of it, either due to space constraints or the assumption that readers already have a clear understanding – though this explanation is less satisfactory for the "Incorrect" category. Because papers go through several rounds of revision before they are eventually published, the incorrect descriptions of the parallel trends assumption that show up in peer-reviewed publications are likely just the tip of the iceberg, a signal of potentially larger issues in working papers, dissertations, seminars, and classrooms. Moreover, any inaccuracies in published papers are likely to be quite consequential as they can perpetuate further misunderstandings. These papers are widely read by those with less experience with econometric methods, including graduate students (in economics or other fields), faculty members outside of the economics field, and those in policy circles, and based on citation counts, poor communication does not appear to lead to lower impact.

References

Baker, A., Callaway, B., Cunningham, S., Goodman-Bacon, A., and Sant'Anna, P. H. (2025). Difference-in-differences designs: A practitioner's guide. *arXiv preprint arXiv:2503.13323*.

Bilinski, A. and Hatfield, L. (2023). Nothing to see here? a non-inferiority approach to parallel trends. *Health Services Research / Working paper*. arXiv:1805.03273.

Borusyak, K., Jaravel, X., and Spiess, J. (2024). Revisiting event study designs: Robust and efficient estimation. *Review of Economic Studies*. Forthcoming / Working paper version January 2024.

Brodeur, A., Mikola, D., Cook, N., Fiala, L., Brailey, T., Briggs, R., de Gendre, A., Dupraz, Y., Gabani, J., Gauriot, R., et al. (2026). Reproducibility and robustness of economics and political science research. *Nature*, 652:151–156.

Caetano, C. and Callaway, B. (2023). Difference-in-differences with time-varying covariates in the parallel trends assumption. *Journal of Econometrics*. Working paper / arXiv:2202.02903.

Callaway, B., Goodman-Bacon, A., and Sant'Anna, P. H. (2024). Difference-in-differences with a continuous treatment. Technical report, National Bureau of Economic Research.

Callaway, B. and Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

Chang, A. C., Li, P., et al. (2018). Is economics research replicable? sixty published papers from thirteen journals say "often not". *Critical Finance Review*, 7:1–25.

de Chaisemartin, C. and D'Haultfœuille, X. (2023). Difference-in-differences estimators of intertemporal treatment effects. *Review of Economic Studies*. Working paper / updated December 2023.

Deshpande, M. (2016). The effect of disability payments on household earnings and income: Evidence from the ssi children's program. *Review of Economics and Statistics*, 98(4):638–654.

Freyaldenhoven, S., Hansen, C., and Shapiro, J. M. (2019). Pre-event trends in the panel event-study design. *American Economic Review*, 109(9):3307–3338.

Garg, P. and Fetzer, T. (2026). Causal claims in economics. Version 2, 24 February 2026.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.

Hengel, E. (2022). Publishing while female: Are women held to higher standards? evidence from peer review. *The Economic Journal*, 132(648):2951–2991.

Huntington-Klein, N., Arenas, A., Beam, E., Bertoni, M., Bloem, J. R., Burli, P., Chen, N., Grieco, P., Ekpe, G., Pugatch, T., et al. (2021). The influence of hidden researcher decisions in applied microeconomics. *Economic Inquiry*, 59(3):944–960.

Isch, C., Dörr, T., Fasching, N., Jennings, G., and Watts, D. J. (2026). Quantifying the prevalence and impact of overreaching causal claims in social science. *Nature Human Behaviour*. Published online 24 August 2026.

Kahn-Lang, A. and Lang, K. (2020). The promise and pitfalls of differences-in-differences: Reflections on 16 and pregnant and other applications. *Journal of Business & Economic Statistics*, 38(3):613–620.

Kosnik, L.-R. (2022). Who are the more dismal economists? gender and language in academic economics research. In *AEA Papers and Proceedings*, volume 112, pages 592–596. American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203.

McCannon, B. C. (2019). Readability and research impact. *Economics Letters*, 180:76–79.

Pütz, P. and Bruns, S. B. (2021). The (non-)significance of reporting errors in economics: Evidence from three top journals. *Journal of Economic Surveys*, 35(1):348–373.

Rambachan, A. and Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591.

Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–322.

Roth, J., Sant'Anna, P. H., Bilinski, A., and Poe, J. (2023). What's trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2):2218–2244.

Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.

A Appendix Tables and Figures

Table A1: Correlates of LLM-Based Paper Classification

	(1) Correct (LLM)	(2) Misinterpretable (LLM)	(3) Incorrect (LLM)
2 authors	-0.04	0.02	0.03
	(0.07)	(0.07)	(0.04)
3 authors	-0.07	0.02	0.04
	(0.08)	(0.08)	(0.04)
4 or more authors	-0.19**	0.12	0.07
	(0.09)	(0.09)	(0.05)
Article Length	-0.05	0.05	-0.00
	(0.03)	(0.03)	(0.02)
Article Length^2	-0.02	0.02	-0.01
	(0.02)	(0.02)	(0.01)
N References	0.00	0.02	-0.02
	(0.04)	(0.04)	(0.02)
N References^2	-0.02	0.01	0.01
	(0.02)	(0.02)	(0.01)
N Incorrect (LLM) Refs	-0.03	0.03	-0.01
	(0.02)	(0.02)	(0.01)
N Misinterpretable (LLM) Refs	-0.03	0.05*	-0.01
	(0.03)	(0.03)	(0.01)
N Correct (LLM) Refs	0.03	-0.03	0.00
	(0.03)	(0.03)	(0.01)
Any New DiD Refs	0.12*	-0.07	-0.05
	(0.06)	(0.06)	(0.03)
Readability	0.01	0.02	-0.03**
	(0.03)	(0.03)	(0.01)
Top 20 Department	-0.11*	0.07	0.03
	(0.06)	(0.06)	(0.03)
Observations	335	335	335
Outcome mean	0.60	0.33	0.07

Robust standard errors reported in parentheses (* p< 0.1, ** p< 0.05, *** p< 0.01). Paper categorization definitions described in Figure 1. All regressions control for year and journal fixed effects.

Table A2: Citations and LLM-Based Paper Classification

	(1) N Citations	(2) N Citations	(3) N Citations	(4) Citations ≥ 50th perc.	(5) Citations ≥ 50th perc.	(6) Citations ≥ 50th perc.	(7) Citations ≥ 75th perc.	(8) Citations ≥ 75th perc.	(9) Citations ≥ 75th perc.
Misinterpretable (LLM)	0.19	-0.01	-0.03	0.03	0.01	0.01	0.01	-0.02	-0.03
	(0.27)	(0.16)	(0.15)	(0.06)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Incorrect (LLM)	0.39	-0.17	-0.39**	0.05	0.01	-0.00	0.02	-0.05	-0.08
	(0.47)	(0.28)	(0.19)	(0.11)	(0.10)	(0.10)	(0.10)	(0.07)	(0.08)
2 authors			0.35**			0.02			0.02
			(0.17)			(0.06)			(0.05)
3 authors			0.04			-0.08			0.00
			(0.21)			(0.07)			(0.06)
4 authors			0.42**			0.03			0.09
			(0.21)			(0.08)			(0.07)
Article Length			-0.02			0.03			-0.01
			(0.08)			(0.03)			(0.02)
Article Length^2			-0.09**			-0.02			-0.01
			(0.04)			(0.01)			(0.01)
N References			-0.14			-0.07**			0.01
			(0.11)			(0.03)			(0.03)
N References^2			0.10***			0.01			0.00
			(0.03)			(0.01)			(0.01)
N Incorrect (LLM) Refs			-0.04			-0.02			0.02
			(0.04)			(0.03)			(0.03)
N Misinterpretable (LLM) Refs			0.10			0.03			0.00
			(0.08)			(0.03)			(0.02)
N Correct (LLM) Refs			-0.03			0.03			0.02
			(0.08)			(0.03)			(0.02)
Any New DiD Refs			0.10			0.10*			-0.01
			(0.21)			(0.05)			(0.05)
Readability			0.06			0.01			0.00
			(0.06)			(0.02)			(0.02)
Top 20 Department			0.28*			0.11**			0.10**
			(0.16)			(0.06)			(0.05)
Observations	335	335	335	335	335	335	335	335	335
Outcome mean	59.57	59.57	59.57	0.50	0.50	0.50	0.25	0.25	0.25
Fixed Effects	None	Year and Journal	Year and Journal	None	Year and Journal	Year and Journal	None	Year and Journal	Year and Journal

Robust standard errors reported in parentheses (* p< 0.1, ** p< 0.05, *** p< 0.01). Paper categorization definitions described in Figure 1. Columns 1 to 3 estimated using PPML.